

Collections Copilot

Helping collections employees prepare a case and respond to customers

A unified customer view, relevant policy and suggested assistance

Team 5C2. Project overview for rubric discussion.

Four connected layers

- 1 Data foundation: match identities across five systems, validate records and build a customer C360 with explicit missingness.
- 2 Questions: answer supported collections questions using trusted SQL, current policy or call evidence, with sources and refusals.
- 3 Features: serve 12 versioned case features with the same offline and online definitions, plus text signals.
- 4 Employee assistance: suggest a response or next step, explain its evidence and record accept, edit or reject decisions.

Six use cases

Agent Assist

Help during the conversation, including hardship, disputes and policy guidance.

Next Best Action

Explain action reasons, alternatives and programme eligibility prerequisites.

Channel comparison

Show outcome support, uncertainty, timing and unmet operational checks.

Agent routing

Match all required skills and language; explain capacity and roster exclusions.

Post-call summary

Separate proposed terms, explicit commitments and follow-ups with timestamps.

Quality assurance

Review context for ten checklist items and record employee evidence per item.

Local reviewed workflows. Treatment effectiveness and production readiness remain unverified.

AI and model choices

Team-trained model: a small text classifier trained on approved notes and customer transcript text to identify financial hardship.

Local models: Whisper transcribes audio. A team-trained Qwen LoRA selects cited case facts. Both run locally without paid API calls.

Optional paid API: an adapter is implemented, but needs user credentials and cost approval. It remains untested.

Controls: deterministic contact gates and employee review remain in place for every model approach.

Current development evidence

130 automated checks pass. The proposal safeguard audit checks 719 approved cases with 0 failures of tested constraints.

All 12 numeric development benchmarks and 24 source-grounded policy/scope checks pass. Broad semantic accuracy remains unmeasured.

Hardship, 181 development examples: classifier 98.6% precision/recall. Live combination: 94.7% precision, 100% recall.

Whisper small.en: 11.02% word error rate on 1803 approved synthetic recordings. Speaker review remains required.

LoRA: 1529 training examples; 468/468 final exact-value outputs. Narrow synthetic task.

Hidden-test results and preparation-time savings remain unmeasured.

Possible grading dimensions

Data correctness: matching quality, financial totals, quarantine, coverage and a reproducible C360 contract.

Question accuracy: benchmark tolerance, usable SQL or source evidence, policy applicability and correct refusals.

Model quality: separate training and evaluation, precision/recall, transcription error and factual draft quality.

Workflow quality: depth of the chosen use case, usability, response time and useful explanations.

Safeguards and reproducibility: consent, holds, privacy, employee control, executable code and honest limitations.

Organiser decisions: paid APIs, pretrained models, use-case breadth and scoring weights.